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Momentum Gap and Go

– Exploring a stock day trading strategy with statistical tools



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- Exploring a stock day trading strategy with statistical tools

In the dynamic and unpredictable stock market, traders face the challenge of capitalizing on rapid price fluctuations. The Momentum Gap and Go strategy has gained prominence among day traders for targeting stocks with significant opening price gaps and high trading volumes. This study investigates the strategy's effectiveness, evaluating its potential profitability and identifying whether the strategy carries any merit.

Using data from 374 small-cap stocks on the New York Stock Exchange (NYSE) over a year, the research analyses stocks with price gaps of 20% or more under varying market conditions and gap sizes. Statistical methods, including correlation and cluster analyses, reveal key relationships between gap size, trading volume, and intraday profitability. Findings suggest smaller gaps (20%-50%) have the highest success rates, with bullish markets favoring specific long and short positions, while bearish markets generally benefit short-selling regardless of gap direction.

The study underscores the strategy's potential for day traders. By providing actionable insights, this research contributes valuable knowledge to the evolving field of day trading strategies.

Keywords: Momentum Trading, Day Trading Strategies, Technical Analysis, Stock Market Gaps, Small-Cap Stocks

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List of abbreviations (or) symbols

1 Introduction

The stock market can be intricate and unpredictable, with companies coming together to trade stocks at fluctuating prices influenced by many factors. Active investors face the task of recognizing and using these price changes to their advantage to gain profit (Raza, S., & Kay-Khine, P. 2023). Among the plethora of strategies utilized by day traders, the Momentum Gap and Go strategy has attracted significant attention (Fernández-Rodríguez et al., 2003). A recent search found a wide variety of social media influencers claiming the high rewards of this specific strategy. This paper aims to find statistical evidence of the claims to see a profit for the momentum gap and go strategy.

1.1 Momentum Gap and Go

The Momentum Gap and Go strategy, a cornerstone of day trading, capitalizes on stocks that exhibit significant price gaps—either upward or downward—accompanied by high trading volumes at the market's opening (Brock et al., 1992). Traders adopting this strategy seek to take positions as the stock price overtakes the opening range, riding the momentum of the prevailing trend (Jegadeesh & Titman, 1993). While this approach can effectively capture short-term price movements, it carries significant risks due to the potential for rapid and unpredictable market shifts (Lo & MacKinlay, 1990).

1.2 Objectives

This thesis examines the factors influencing the success of the Momentum Gap and Go day trading strategy from the stock market's opening to the closing bell. By analyzing extensive datasets of stock prices and trading volumes, we strive to identify patterns and trends that might support and equip traders with actionable intelligence for this strategy.

Does statistical evidence support the day trading Momentum Gap and Go strategy for investment success? The goal of this paper is to answer this question with suggestions on what aspects are indicators of finding this success.

In this introductory chapter, we have outlined the central theme of our thesis and provided foundational information on the Momentum Gap and Go day trading strategy. The later chapters will delve deeper into this topic, starting with a comprehensive literature review, followed by a detailed explanation of our research methods, the presentation of our findings, and a discussion of the implications these findings have for the broader context of stock market trading. Our goal is to contribute meaningfully to the existing knowledge of this widely used day trading technique and offer valuable insights that can enhance traders' performance.

In this introductory chapter, we have outlined the central theme of our thesis and provided foundational information on the Momentum Gap and Go day trading strategy. The subsequent chapters will delve deeper into this topic, starting with a comprehensive literature review, followed by a detailed explanation of our research methods, the presentation of our findings, and a discussion of the implications these findings have for the broader context of stock market trading. Our goal is to contribute meaningfully to the existing knowledge of this widely used day trading technique and offer valuable insights that can enhance traders' performance.

2 Technical vs Fundamental Analysis

Technical and fundamental analysis are two common, distinct approaches to evaluating financial markets and making investment decisions. Fundamental analysis examines a company's intrinsic value by analyzing financial statements, economic factors, and industry conditions to determine if a security is under or overvalued (Kirkpatrick & Dahlquist, 2010). In contrast, technical analysis studies historical price movements and trading volumes to identify patterns and trends, assuming that past market behavior can predict future price action (Chowdhury, 2021). While fundamental analysis is typically associated with long-term investment strategies, technical analysis is often employed for short-term trading decisions (Abdullah et al., 2022).

The debate between the efficacy of technical and fundamental analysis has been ongoing in academic literature. Some studies suggest combining both approaches may lead to more informed investment decisions (Chowdhury et al., 2022). Fundamental analysis provides a comprehensive understanding of a company's financial health and growth potential, which is crucial for long-term investors (Francis, 1998). However, technical analysis can offer insights into market sentiment and timing, which may be valuable for short-term traders or for optimizing entry and exit points (Kirkpatrick & Dahlquist, 2010). The choice between these methods often depends on an investor's goals, time horizon, and risk tolerance, with some practitioners advocating for a blended approach to capitalize on the strengths of both methodologies (Chowdhury, 2021).

The questions and focus of this study are specific to the technical analysis of market price action. Future studies can use this paper's technical analysis along with other fundamental analyses for a more long-term perspective.

2.1 Technical Indicators

Technical analysis, used for forecasting price movements in financial markets, relies on several critical data types to evaluate market conditions. This section examines the primary raw data sources utilized in technical analysis and their significance in market evaluation.

2.2 Price Data

Price data forms the cornerstone of technical analysis. It typically encompasses the open, high, low, and close (OHLC) prices for each period, ranging from minute-by-minute intervals to daily or weekly timeframes (Murphy, 1999). This data serves as the foundation for most technical indicators and chart patterns. For instance, candlestick charts, widely employed in technical analysis, are constructed using OHLC price data. The importance of price data cannot be overstated in technical analysis, as it directly reflects market valuation and serves as the primary input for trend analysis and pattern recognition.

2.3 Volume Data

Volume data, representing the number of contracts or shares traded during a specific period, is a crucial complement to price data in technical analysis. It provides insight into the strength of price movements and market participation (Kirkpatrick & Dahlquist, 2010). Analysts often use volume and price data to confirm trends or identify potential reversals. High volume accompanying a price move is generally interpreted as a strong confirmation of the move, while low volume may suggest a lack of trust in the market.

2.4 Time Data

The timestamp associated with each price and volume data point is essential for technical analysis. It allows analysts to create time-based charts and indicators, such as moving averages or oscillators, which are fundamental to many technical analysis strategies (Murphy, 1999). Time data enables examining

market behavior over different periods, facilitating short-term and long-term analysis.

In conclusion, these raw data types form the foundation of technical analysis. The data is often collected in real-time or near-real-time from exchanges and data providers, then processed and analyzed using various technical indicators and charting tools. Interpreting these data points, individually and in combination, allows technical analysts to predict future price movements and market trends. Using these raw data sources effectively is crucial for developing robust technical analysis strategies and making informed trading decisions.

3 Literature Review

In previous chapters, we introduced the Momentum Gap and Go day trading strategy, noting its prevalence among active investors who use price movements in stocks that show significant gaps at the market's opening with high trading volume. This strategy is predicated on the notion that stocks experiencing substantial price changes at the opening bell are likely to maintain that momentum in the short term (Brock et al., 1992; Jegadeesh & Titman, 1993).

3.1 Momentum

Much like an ocean wave will predictably move in a predetermined direction by the momentum of its mass, the logic is similarly applied to the financial movements of a heavily moving stock price momentum. According to Fidelity.com, momentum is the speed at which the mass of price movement progresses. This study will focus on the significant momentum gaps from the previous market closing price to the next-day market opening price. A considerable gap will be defined as anything above a 20% increase or decrease in price movement from the previous day's close.

3.2 Bear Markets vs. Bull Markets

General market trends describe the movement of the market's price action. According to Wohlner, R. (2023), bear markets are an average trend of the market prices and market sentiment moving downwards, or losing money, on the stock prices. In contrast, the bull market refers to the market prices and market sentiment moving upwards or gaining money from stock prices. These prevailing market conditions influence the efficiency of the momentum gap and go strategy. Lo and MacKinlay (1990) found that this strategy yields better outcomes during bull markets than bear markets. In line with this, Fernández-

Rodríguez, Sosvilla-Rivero, and Andrada-Félix (2003) observed an increase in the strategy's effectiveness amidst heightened market volatility.

3.3 Small-Cap vs. Large-Cap

Stock characteristics also play a pivotal role in the strategy's success.

Jegadeesh and Titman (1993) discovered that the strategy is more successful with small-cap stocks than large-cap alternatives. According to Wayman, R. (2022), small-cap stocks are defined to have a market value cap between \$250 million and \$2 billion; for this study, we will focus on stocks that fit within this category. Similarly, Brock, Lakonishok, and LeBaron (1992) indicated that stocks with lower price-to-earnings ratios tend to be more conducive to this strategy.

3.4 Volume

Trading volume is another critical factor that influences the outcomes of the Momentum Gap and Go strategy. Lo and MacKinlay (1990) noted that high trading volumes are associated with better strategy performance. Fernández-Rodríguez, Sosvilla-Rivero, and Andrada-Félix (2003) found that a significant imbalance between buy and sell orders enhances the strategy's effectiveness.

3.5 Additional Insight

Despite these valuable insights, there still needs to be a more comprehensive understanding of how market conditions, stock characteristics, and trading volume interact to determine the performance of the Momentum Gap and Go strategy. The interaction with each described trading characteristic is a stepping stone to a comprehensive understanding of the ebb and flow of price movements on the high fluctuation of price actions within the market. Further research is essential to unravel the complexities involved and to provide a more

holistic understanding of these factors' collective impact on the strategy's effectiveness.

The following chapters of this thesis will present empirical research methods and findings, aiming to contribute to this expanding body of knowledge. By analyzing real-world data, we intend to illuminate factors that influence the success of the Momentum Gap and Go day trading strategy, providing practical insights for traders looking to refine their techniques.

4 Methodology

This chapter outlines the methodological framework for collecting, processing, and analyzing data to address the research question: "What factors influence the success of the Momentum Gap and Go day trading strategy during the stock market's opening to the closing bell?"

Parameters used for the study:

- Date – 19/11/2023 – 18/11/2024
- Bull vs. Bear Market trend – A 10-day moving average with a price up is a “Bull” market, and a 10-day moving average with a price down is a “Bear” market.
- Gap size – greater than 20% price movement both up and down in price
- Historical data – sourced from [yahoo.finance.com](https://finance.yahoo.com)

4.1 Data Collection

The dataset was compiled by scraping a comprehensive list of small-cap stocks traded on the New York Stock Exchange (NYSE) from StockAnalysis.com. The primary focus is stocks exhibiting significant price gaps ($\geq 20\%$) from the previous day's closing price. The data covered one year and included variables such as:

- Opening and closing prices.
- Daily highs and lows.
- Trading volume.
- Prevailing market conditions (bull or bear).

4.2 Data Gathering and Preparation

A Python script is used to pull out data from Yahoo.finance.com; for the entire script, refer to Appendix 2. As seen in Figure 1, the data is collected from historical data cleaned out within the past year, given a parameter or threshold to find stocks with a minimum and maximum stock price gap.

```
def analyze_stock(tickers, gap_min_threshold=20, gap_max_threshold=50,
output_file='stock_analysis20.xlsx'):
    results = []

    for ticker in tickers:
        print(f"Processing ticker: {ticker}")

        try:
            # Download historical data for the past year
            stock = yf.Ticker(ticker)

            # Custom date range for the past year
            end_date = datetime.today()
            start_date = end_date - timedelta(days=365)
            data = stock.history(start=start_date, end=end_date)

            if data.empty:
                print(f"No data found for ticker {ticker}.
Skipping.\n")
                continue
```

Figure 1.

Next, the data is processed to determine the percentage of the gap difference between the previous market close and the current day's opening price (Figure 2). Sorting out the necessary data that is in the interest of this study.

```

data.reset_index(inplace=True) # Reset index to include
'Date'

# Add previous close column
data['Prev_Close'] = data['Close'].shift(1)
data.dropna(inplace=True) # Remove rows with NaN

# Calculate gap percentage
data['Gap_Percent'] = ((data['Open'] - data['Prev_Close'])
/ data['Prev_Close']) * 100

# Filter rows where gap is between the min and max
threshold
gap_data = data[(data['Gap_Percent'].abs() >=
gap_min_threshold) & (data['Gap_Percent'].abs() <=
gap_max_threshold)].copy()

if gap_data.empty:
    print(f"No gap events found for ticker {ticker} within
the range {gap_min_threshold}% to {gap_max_threshold}%. Skipping.\n")
    continue

```

Figure 2.

Following along, as seen in Figure 3, the script determines whether there is a “Bear” or “Bull” market alongside the desired data points. A 10-day historical average of the stock in question determines this.

```

# Determine bull or bear market
data['Trend'] =
data['Close'].rolling(window=10).mean().diff()
bull_bear = "Bull" if data['Trend'].iloc[-1] > 0 else
"Bear"

```

Figure 3.

Figure 4 shows the process of inserting the labels of the rows of the desired data points.

```
# Add relevant data to results
for _, row in gap_data.iterrows():
    results.append({
        "Ticker": ticker,
        "Date": row['Date'],
        "Open": row['Open'],
        "Close": row['Close'],
        "High": row['High'],
        "Volume": row['Volume'],
        "Prev_Close": row['Prev_Close'],
        "Gap_Percent": row['Gap_Percent'],
        "Market": bull_bear,
    })

except Exception as e:
    print(f"Error processing ticker {ticker}: {e}")
    continue
```

Figure 4.

Finally, to make the script friendly for data analysis, we converted the data into an Excel sheet where the observations for this study can be studied (Figure 5).

```
# Convert results to DataFrame
results_df = pd.DataFrame(results)

if not results_df.empty:
    # Ensure the Date column is timezone-unaware
    results_df['Date'] = results_df['Date'].dt.tz_localize(None)

    # Clean out rows where 'Open' price is 0
    results_df = results_df[results_df['Open'] != 0]

    # Sort by date
    results_df.sort_values(by='Date', inplace=True)

    # Save to Excel
    results_df.to_excel(output_file, index=False)
    print(f"Analysis saved to {output_file}")
else:
    print(f"No valid gap events found for the given tickers with a
gap percentage between {gap_min_threshold}% and {gap_max_threshold}%.")

tickers =
pd.read_csv('/content/small_cap_stocks.csv')['Ticker'].tolist()

analyze_stock(tickers)
```

Figure 5.

4.3 Analytical Framework

The analysis employed a multi-faceted statistical approach, integrating the following techniques:

- a. Descriptive Statistics: Summarized key data characteristics, such as gap size distributions, intraday price ranges, and success probabilities by market conditions, as seen in Figure 6.

Ticker	Date	Open	Close	High	Volume	Prev_Close	Gap_Percent	Market
EYPT	04/12/2023	22	18,34000015	22,44000053	20827300	6,610000134	232,83	Bear
CSR	11/03/2024	56,40999985	56,56000137	56,79000092	37100	20,39999962	176,52	Bull
LENZ	22/03/2024	15,97999954	16,20000076	18,19499969	67400	3,180000067	402,52	Bull
MTTR	22/04/2024	4,050000191	4,800000191	4,989999771	72294500	1,740000001	132,76	Bull
AMC	14/05/2024	11,88000011	6,849999905	11,88000011	634246600	5,190000057	128,90	Bull
DSY	10/06/2024	3,940000057	3,609999895	5,139999866	53313100	1,909000039	106,39	Bear
TECX	24/06/2024	17,20000076	16,10000038	17,20000076	43500	1,399999976	1128,57	Bear

Figure 6.

- b. Correlation Analysis: Explored relationships between variables like trading volume and closing price, revealing that volume had minimal impact on end-of-day profitability.
- c. Cluster Analysis: Segmented stocks into groups based on momentum gap size, allowing identification of trends, such as higher success rates for smaller gaps (20%-50%).
- d. Market-Specific Analysis: Examined strategy performance in bull and bear markets. Observations highlighted that:

Bear Market	Gap up	Gap Down
Number of Gains	50	39
Number of Losses	59	44
Percentage of wins	0,458715596	0,469879518
Percentage of loss	0,541284404	0,530120482

Figure 7.

- o Bear markets slightly favor short positions regardless of gap direction (See Figure 7).

Bull Market	Gap up	Gap Down
Number of Gains	52	47
Number of Losses	61	22
Percentage of Wins	0,460176991	0,68115942
Percentage of Loss	0,539823009	0,31884058

Figure 8.

- Bull markets favor long positions on downward gaps and short positions on upward gaps (See Figure 8).

4.4 Methodological Rigor

This rigorous, data-driven methodology ensured a comprehensive examination of the factors influencing the success of the Momentum Gap and Go strategy. The next chapter presents detailed results from these analyses, offering more profound insights into the strategy's practical implications.

5 Observed Data

This chapter presents a detailed analysis of the gathered and observed data, highlighting trends and relationships influencing the Momentum Gap and Go strategy. The data (see Appendix 1) is cleaned out from 1428 stocks in 252 trading days, totaling 374 small-cap stocks traded on the New York Stock Exchange (NYSE) in the past year that experienced a minimum price gap of 20% from the previous day's closing price.

5.1 Overview of Gapped Stocks

Of the 374 stocks analyzed:

- **nine stocks** exhibited price gaps exceeding 100%.
- **33 stocks** displayed price gaps between 50% and 100%.
- **332 stocks** showed price gaps between 20% and 50%.

The dataset was divided into three groups based on gap size:

- **Group 1:** Gaps greater than 100%.
- **Group 2:** Gaps between 50% and 100%.
- **Group 3:** Gaps between 20% and 50%.

The average probability of achieving a positive financial outcome varied across these groups:

- Group 1: **44.44%**
- Group 2: **42.42%**
- Group 3: **49.70%**

These findings suggest that smaller gaps (20%-50%) may yield a slightly higher likelihood of profitability within a single trading day.

5.2 Volume Analysis

The analysis revealed no significant correlation between trading volume and closing price within the one-day timeframe of the Momentum Gap and Go strategy. While trade volume may play a more substantial role in shorter-term intraday movements, its impact on end-of-day profitability for this strategy is limited.

5.3 Price Range Observations

As found in research displayed in Appendix 1, for stocks experiencing a gap of 20% or more, the mean intraday price range (high minus low) was **\$1.76**. The most extensive observed intraday price range was a striking **\$146.46**, reflecting extreme volatility in specific stocks. These figures underscore the importance of price range when implementing the strategy.

5.4 Bull Market Observations

In a Bull market:

- **Price gaps down** of 20% or more resulted in a **68.12% probability** of ending the day in profit when taking a long position.
- **Price gaps up** to 20% or more resulted in a **53.98% probability** of a price decline by market close.

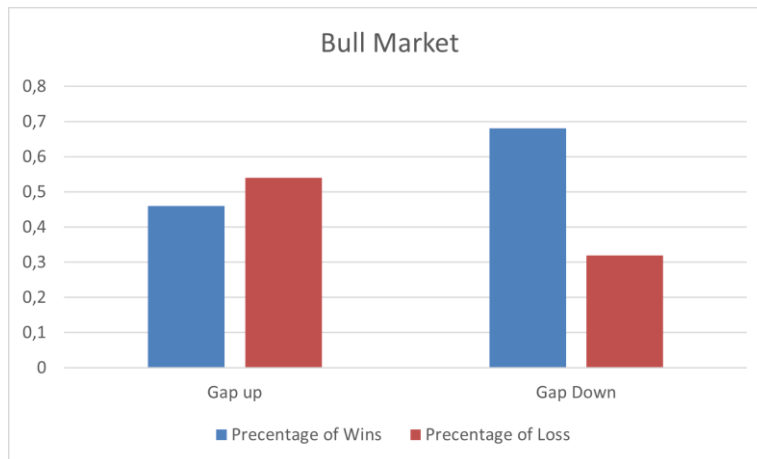


Figure 9.

Figure 1 shows that in bullish market conditions, a long position following a downward gap and a short position following an upward gap are statistically favorable strategies.

5.5 Bear Market Observations

In a Bear market:

- **Price gaps down** of 20% or more had a **53.01% probability** of further price decline by the end of the day.
- **Price gaps up** to 20% or more had a **54.13% probability** of a price decline by market close.

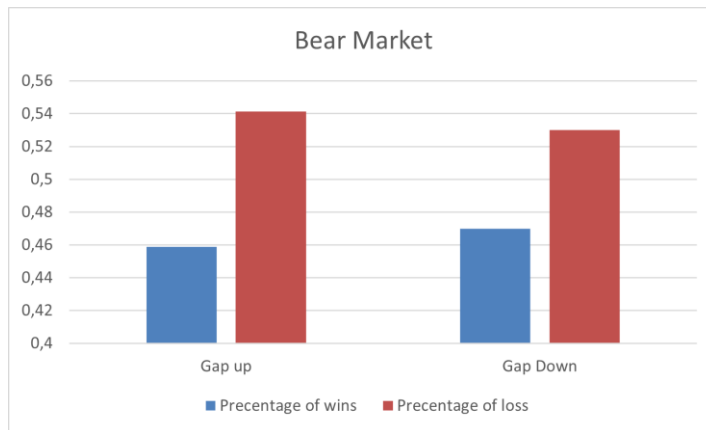


Figure 10.

In bearish market conditions, as seen in Figure 2, short positions, regardless of the direction of the initial gap, tend to be slightly more likely to be profitable.

6 Conclusion

This study has explored the *Momentum Gap and Go* strategy as a tool for technical day trading, emphasizing its application under specific market conditions. Through a systematic analysis of historical data, we identified specific favorable scenarios, such as the higher profitability of long positions during downward gaps in bullish markets and short positions in upward gaps across varying market conditions. These insights contribute to a better understanding of how momentum-based strategies can be applied effectively in day trading.

While this research has primarily relied on technical analysis, future studies may consider the incorporation of fundamental analysis. This would provide a more robust decision-making framework, enabling traders to assess market sentiment and broader economic factors, enhancing the strategy's effectiveness.

As a natural progression from this research, future studies should be guided by a thorough understanding of the limitations and complexities inherent in the Momentum Gap and Go strategy. This will ensure that the research is focused on areas that can significantly improve the strategy's effectiveness. The following are suggestions for future consideration, in no order of significance.

1. Risk Analysis: Incorporating a comprehensive risk assessment framework to evaluate the downside potential of the strategy, particularly during high volatility periods.
2. High-Frequency Data: Analyzing intraday price movements at smaller increments (minutes and seconds) to better understand price action dynamics and refine entry/exit strategies.
3. Automated Trading Systems: Investigating how algorithmic trading systems could implement the strategy.
4. Market Sentiment and Fundamentals: Expanding the scope of analysis to include fundamental indicators such as earnings reports, news

sentiment, and macroeconomic data, potentially offering insights to technical patterns.

5. Comparative Analysis: Examining the strategy across different market sectors, international exchanges, or during extreme market events to evaluate its generalizability and robustness.

The findings of this study offer valuable contributions to the domain of day trading strategies while acknowledging the limitations inherent in a single-day, technical analysis-focused framework. As the financial markets continue to evolve with advancements in technology and trading algorithms, the potential for refining and expanding the *Momentum Gap and Go* strategy is notable.

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Scraped and Cleaned Raw Stock Example Data

Ticker	Date	Open	Close	High	Volume	Prev Close	Gap	Percent	Market	Low	Days Price Range	Potential Long P Gain	Potential Short P Gain	End of Day	Gain or Loss
EYPT	04/12/2023	22	18,34800015	22,44000015	20827900	6,510000134	232,83	Bear	17,31	5,13	0,44	4,89	3,10	1	1
CSR	11/03/2024	56,40999985	56,56000137	56,79000092	37100	20,39999962	176,52	Bull	56,24	0,55	0,38	0,17	-0,15	-1	-1
LENZ	22/03/2024	15,97999954	16,20000076	18,19499969	67400	3,180000067	402,52	Bull	15,09	3,10	2,22	0,89	-0,22	-1	-1
MTRT	22/04/2024	4,050000191	4,800000191	4,989999771	72294500	1,740000001	132,76	Bull	3,24	1,75	0,94	0,81	-0,75	-1	-1
AMC	14/05/2024	11,88000011	6,849999905	11,88000011	634246600	5,190000057	128,90	Bull	5,85	6,03	0,00	6,03	5,03	1	1
DSY	10/06/2024	1,940000057	3,609999895	5,139999886	53313100	1,909000039	106,39	Bear	3,40	1,74	1,20	0,54	0,33	1	1
TECK	24/06/2024	17,20000076	16,10000038	17,20000076	43500	1,399999916	1128,57	Bear	15,47	0,00	1,73	1,73	1,10	1	1
NAPA	07/10/2024	10,89000034	10,94999981	11	46907100	5,400000095	101,67	Bull	10,86	0,14	0,11	0,03	-0,06	-1	-1
ROOT	31/10/2024	81,48000036	68,38999939	118,1500015	13509200	40,49000168	101,23	Bull	63,12	55,03	36,67	18,36	13,09	1	1
ALT	01/12/2023	5,739999771	3,569999933	5,869999886	100948400	3,160000086	81,65	Bull	3,52	2,35	0,13	2,22	2,17	1	1
REPL	05/12/2023	6,099999905	6,75	7,230000019	13908700	12,319999969	-50,49	Bear	5,89	1,34	1,13	0,21	-0,65	-1	-1
COST	11/12/2023	4,289999962	4,059999962	4,590000917	20991900	8,640000343	-60,35	Bear	3,67	0,92	0,30	0,62	0,23	1	1
CCCC	12/12/2023	1,820000052	2,339999914	2,700000048	179486300	1,179999948	54,24	Bear	1,76	0,94	0,06	0,88	-0,52	-1	-1
ZVL	19/12/2023	8,75	6,539999767	9,5	6196000	5,093500137	71,79	Bear	5,19	4,31	0,75	3,56	3,06	1	1
FDMT	05/02/2024	30,04999924	32,79000092	35,61299986	19637700	17,48999977	71,81	Bull	28,75	6,86	5,56	1,30	-2,24	-1	-1
ENGN	14/02/2024	13,75	16,5	16,80999947	867200	7,619999886	80,45	Bear	12,08	4,73	3,06	1,67	-2,75	-1	-1
COST	02/03/2024	8,550000191	8,699999809	9,414999962	10707800	5,460000038	56,59	Bear	7,95	1,45	0,86	0,60	-0,15	-1	-1
NWOK	15/02/2024	12,42999958	9,5	12,69999981	40777300	6,360000134	95,60	Bull	9,19	3,51	0,26	3,25	2,94	1	1
SOUN	15/02/2024	4,010000229	3,759999999	4,059999943	298289400	2,255000114	77,83	Bull	3,11	0,95	0,05	0,90	0,25	1	1
BYND	28/02/2024	11,30000019	9,829999924	12,119999989	44768200	7,519999981	50,27	Bear	9,62	2,50	0,82	1,68	1,47	1	1
FUFU	04/03/2024	12,28999996	13,13000011	18,139999969	7585700	6,380000114	92,63	Bull	11,90	6,42	6,03	0,39	-0,84	-1	-1
CSR	06/03/2024	19,49311025	19,55103874	19,59931218	50700	54,07672119	-63,95	Bull	55,38	35,78	0,11	-35,88	-0,06	-1	-1
STOK	26/03/2024	11,06999969	10,28999996	13	12468000	6,5	70,16	Bear	10,36	1,99	1,03	0,91	0,78	1	1
IRON	01/04/2024	26,5	34,20999908	35,40000153	6218200	62,25999832	-57,44	Bear	25,60	9,80	8,90	0,90	-7,71	-1	-1
ORKA	03/04/2024	33,5750632	37,497248	44,76675463	6734208	19,7296772	70,18	Bear	2,88	41,89	11,19	30,70	-3,92	-1	-1
CLYM	11/04/2024	4,010000229	4,579999924	4,75	39106000	2,670000076	50,19	Bear	3,92	0,83	0,74	0,09	-0,57	-1	-1
EB5	02/05/2024	2,950000048	3,299999952	4,760000229	151505000	1,529999948	52,85	Bear	2,82	1,94	1,81	0,13	-0,35	-1	-1
TMCI	08/05/2024	4,5	4,70000076	4,639999866	15641400	11,119999988	-50,53	Bull	3,92	0,72	0,14	0,58	0,13	1	1
PLUG	14/05/2024	4,889999866	3,440000057	4,900000095	203418400	2,880000105	69,20	Bear	3,42	1,48	0,01	1,47	1,45	1	1
CALT	28/05/2024	38,15000153	37,95000076	38,31999969	218781	22,42000008	70,16	Bull	5,22	33,10	0,17	32,93	0,20	1	1
BMEA	07/06/2024	3,880000114	4,199999866	4,349999905	11882800	11,270000046	-65,57	Bear	3,61	0,74	0,07	0,27	-0,26	-1	-1
SHCR	21/06/2024	1,360000014	1,370000005	1,379999995	56333782	0,774399996	75,62	Bull	1,19	0,42	1,36	-0,01	-1	-1	
TECK	01/06/2024	1,399999976	1,399999976	1,399999976	0	16,79999924	-91,65	Bear	16,05	14,65	0,00	-14,65	0,00	-1	-1
SERV	19/07/2024	4,269999991	7,550000191	8,970000267	266271300	2,630000114	62,36	Bull	5,02	4,70	0,35	4,35	-2,28	-1	-1
ITI	09/08/2024	6,989999771	7	7,014999866	23399620	4,280000021	63,32	Bull	6,97	0,04	0,03	0,02	-0,01	-1	-1
RVNC	12/08/2024	6,554999828	6,599999905	6,599999905	61906700	3,529999971	85,69	Bear	6,53	0,07	0,05	0,02	-0,05	-1	-1
APLD	05/09/2024	4,989999771	5,369999886	5,710000038	149889900	3,240000001	54,01	Bull	4,25	1,46	0,72	0,74	-0,38	-1	-1
LUNR	18/09/2024	8,380000114	7,469999979	8,949999809	141864600	5,400000095	55,19	Bull	7,44	1,51	0,57	0,94	0,91	1	1
RZLV	19/09/2024	8,130000114	7	10,76000002	7604800	4,25	71,16	Bear	5,62	2,51	1,13	1,13	1	1	
SUN	23/09/2024	17,58042223	17,62214851	17,69353292	3017491	58,90000153	-70,14	Bear	11,51	6,08	0,94	-0,03	-1	-1	
HEPS	18/10/2024	3,529999971	3,509999999	4,050000191	43863300	2,200000048	60,45	Bear	2,90	1,15	0,52	0,63	0,02	1	1
ANRO	23/10/2024	5,150000095	4,360000134	5,730000019	16354300	14,52999973	-64,56	Bull	4,26	1,47	0,58	0,89	0,79	1	1
LESL	29/11/2023	4,5	5,230000019	5,230000019	14795500	5,820000172	-22,68	Bull	4,46	0,07	0,73	0,04	-0,73	-1	-1
BMEA	29/11/2023	12,11999989	12,75	15,29500008	4441600	9,939999958	70,18	Bull	11,68	3,62	2,19	0,88	-0,63	-1	-1
REV	30/11/2023	48,06000137	49,02000046	51,18999962	290200	36,88999939	29,20	Bear	5,13	3,13	2,00	-0,96	-1	-1	
PHV	04/12/2023	1,669999957	1,539999962	1,690000057	23731500	1,370000005	21,90	Bull	1,46	0,23	0,02	0,21	0,13	1	1
PHVS	06/12/2023	26,87000084	24,75	29,79999924	2227700	20,95000076	28,26	Bear	24,20	5,60	2,92	2,67	2,12	1	1
PHR	06/12/2023	18,15999985	17,87999916	19,03499985	2875400	14,75	23,12	Bull	17,38	1,65	0,88	0,78	0,28	1	1
WVE	07/12/2023	4,630000114	4,78000021	4,949999809	5187900	6,869999886	-32,61	Bear	4,60	0,35	0,32	0,03	-0,15	-1	-1
GBL	07/12/2023	26,389999962	28,75	29,23999997	55000	33,95999939	-20,70	Bull	2,34	2,60	0,40	2,60	-1,85	-1	-1
CXM	07/12/2023	11,98999977	11,10999956	12,10999966	24759000	16,70000076	-28,20	Bull	10,96	1,15	0,12	1,03	0,88	1	1
CURV	08/12/2023	5,25	4,840000153	5,93200016	1757400	4,150000095	26,51	Bull	4,50	1,43	0,68	0,75	0,41	1	1
BMEA	11/12/2023	12,75	11,77000046	12,93999958	7550800	16,88999939	-24,51	Bear	10,37	2,57	0,19	2,39	0,98	1	1
CCCC	13/12/2023	3,180000067	5,510000229	6,03000021	225846300	2,339999914	-35,90	Bear	3,16	2,87	2,85	0,02	-2,33	-1	-1
ESPR	14/12/2023	1,850000014	1,730000019	1,960000038	28059100	1,509999999	22,52	Bull	1,71	0,24	0,11	0,12	0,11	1	1
EOSE	15/12/2023	1,129999995	1,129999995	1,119999986	43212100	1,679999948	-32,74	Bear	1,01	0,13	0,01	0,12	0,00	-1	-1
ZINV	18/12/2023	14,23999977	17,10000038	17,19000053	3518400	11,22000027	26,92	Bear	14,02	3,17	2,95	0,22	-2,86	-1	-1
GOTU	18/12/2023	3,690000057	3,730000019	3,900000095	21283900	4,630000114	-20,30	Bull	3,61	0,30	0,21	0,09	-0,04	-1	-1
ARQT	18/12/2023	3,059999943	2,539999962	3,430000067	45222900	2,440000057	25,41	Bull	2,46	0,97	0,37	0,60	0,52	1	1
MIRM	18/12/2023	3,470000029	2,799999952	3,539999962	23917800	4,960000038	-21,64	Bull	27,37	2,36	0,00	-2,01	-1	-1	
PCY	18/12/2023	3,609999805	4,099999905	4,679999886	16572800	5,820000172	-37,97	Bear	3,41	1,27	1,07	0,20	-0,49	-1	-1
CVAC	19/12/2023	3,463000059	4,20025149	4,89349851	3428000	5,693999767	-39,18	Bear	2,34	2,56	1,43	1,13	-0,74	-1	-1
ZVL	21/12/2023	3,599999905	3,730000019	4,039999962	13565100	2,880000114	25,00	Bull	3,27	0,77	0,44	0,33	-0,13	-1	-1
ZVL	29/12/2023	10,75	10,4250015	13,77700015	1444000	8,699999809	23,56	Bull	10,20	3,58	0,55	0,33	-1,65	-1	-1
AVUL	01/01/2024	9,900000095	6,945000076	6,949999886	9264800	8,310000027	-36,63	Bull	5,79	1,05	0,11	0,11	-0,14	-1	-1
WULF	02/01/2024	2,930000057	2,299999952	2,940000057	7016700	2,400000095	22,08	Bull	0,71	0,01	0,70	0,63	0,1	1	1
VYGR	02/01/2024	10,52999973	9	11,72000027	12027800	8,43999958	24,76	Bear	8,90	2,82	1,19	1,63	1,53	1	1
AGL	05/01/2024	7,75	8,630000114	8,819999695	36877000	12,07999992	-35,84	Bear	7,59	1,23	1,07	0,16	-0,88	-1	-1
APLT	05/01/2024	2,651000023	2,25	2,75999999	6795800	3,789999962	-30,05	Bear	2,23	0,53	0,11	0,42	0,40	1	1
DCGZ	08/01/2024	5,690000057	5,190000057	5,78000021	203500	4,380000114	29,91	Bull	4,44	1,25	0,50	0,83	0,76	1	1
PRTA	08/01/2024	21,54000002	35,79999978	35,54000002	3795300	40,									

Python Script for Data Extraction

```
import yfinance as yf
import pandas as pd
from datetime import datetime, timedelta

def analyze_stock(tickers, gap_min_threshold=20,
gap_max_threshold=50, output_file='stock_analysis20.xlsx'):
    results = []

    for ticker in tickers:
        print(f"Processing ticker: {ticker}")

        try:
            # Download historical data for the past year
            stock = yf.Ticker(ticker)

            # Custom date range for the past year
            end_date = datetime.today()
            start_date = end_date - timedelta(days=365)
            data = stock.history(start=start_date, end=end_date)

            if data.empty:
                print(f"No data found for ticker {ticker}.
Skipping.\n")
                continue

            data.reset_index(inplace=True) # Reset index to
include 'Date'

            # Add previous close column
            data['Prev_Close'] = data['Close'].shift(1)
            data.dropna(inplace=True) # Remove rows with NaN

            # Calculate gap percentage
            data['Gap_Percent'] = ((data['Open'] -
data['Prev_Close']) / data['Prev_Close']) * 100

            # Filter rows where gap is between the min and max
threshold
            gap_data = data[(data['Gap_Percent'].abs() >=
gap_min_threshold) & (data['Gap_Percent'].abs() <=
gap_max_threshold)].copy()

            if gap_data.empty:
```

```

        print(f"No gap events found for ticker {ticker}
within the range {gap_min_threshold}% to {gap_max_threshold}%.
Skipping.\n")

        continue

    # Determine bull or bear market
    data['Trend'] =
data['Close'].rolling(window=10).mean().diff()
    bull_bear = "Bull" if data['Trend'].iloc[-1] > 0 else
"Bear"

    # Add relevant data to results
    for _, row in gap_data.iterrows():
        results.append({
            "Ticker": ticker,
            "Date": row['Date'],
            "Open": row['Open'],
            "Close": row['Close'],
            "High": row['High'],
            "Volume": row['Volume'],
            "Prev_Close": row['Prev_Close'],
            "Gap_Percent": row['Gap_Percent'],
            "Market": bull_bear,
        })

    except Exception as e:
        print(f"Error processing ticker {ticker}: {e}")
        continue

    # Convert results to DataFrame
    results_df = pd.DataFrame(results)

    if not results_df.empty:
        # Ensure the Date column is timezone-unaware
        results_df['Date'] =
results_df['Date'].dt.tz_localize(None)

        # Clean out rows where 'Open' price is 0
        results_df = results_df[results_df['Open'] != 0]

        # Sort by date
        results_df.sort_values(by='Date', inplace=True)

        # Save to Excel
        results_df.to_excel(output_file, index=False)
        print(f"Analysis saved to {output_file}")

```

```
        else:
            print(f"No valid gap events found for the given tickers  
with a gap percentage between {gap_min_threshold}% and  
{gap_max_threshold}%.")

tickers =
pd.read_csv('/content/small_cap_stocks.csv')['Ticker'].tolist()

analyze_stock(tickers)
```